MA1522 Linear Algebra for Computing
L2 Lesson 7



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Linear Combinations and Linear Span

Let $\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_k \in \mathbb{R}^n$.

- A **linear combination** of $\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_k$ is a vector of the form

$$c_1\boldsymbol{u}_1 + c_2\boldsymbol{u}_2 + \cdots + c_k\boldsymbol{u}_k$$

  where $c_1, \ldots, c_k \in \mathbb{R}$.

- The **(linear) span** of $\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_k$, denoted $\operatorname{span}\{\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_k\}$, is the set of all linear combinations of $\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_k$, i.e.

$$\operatorname{span}\{\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_k\} = \{c_1\boldsymbol{u}_1 + c_2\boldsymbol{u}_2 + \cdots + c_k\boldsymbol{u}_k \mid c_1, c_2, \ldots, c_k \in \mathbb{R}\}.$$

For $\boldsymbol{v} \in \mathbb{R}^n$, we have

$$\begin{aligned}
&\boldsymbol{v} \in \operatorname{span}\{\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_k\} \\
&\Leftrightarrow \boldsymbol{v} = c_1\boldsymbol{u}_1 + c_2\boldsymbol{u}_2 + \cdots + c_k\boldsymbol{u}_k \text{ for some } c_1, \ldots, c_k \in \mathbb{R}^n \\
&\Leftrightarrow \boldsymbol{v} = \begin{pmatrix} \boldsymbol{u}_1 & \boldsymbol{u}_2 & \cdots & \boldsymbol{u}_k \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_k \end{pmatrix} \text{ for some } c_1, \ldots, c_k \in \mathbb{R}^n \\
&\Leftrightarrow \boldsymbol{A}\boldsymbol{x} = \boldsymbol{v} \text{ has a solution,}
\end{aligned}$$

where $\boldsymbol{A} = \begin{pmatrix} \boldsymbol{u}_1 & \boldsymbol{u}_2 & \cdots & \boldsymbol{u}_k \end{pmatrix}$.

Question 1

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Let $\boldsymbol{u}_1 = (2, 1, 0, 3)$, $\boldsymbol{u}_2 = (3, -1, 5, 2)$, $\boldsymbol{u}_3 = (-1, 0, 2, 1)$.

Determine whether each of the following vectors belongs to $\text{span}\{\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3\}$, and if so, express it as a linear combination of $\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3$:

a. $\boldsymbol{v}_1 = (2, 3, -7, 3)$,
b. $\boldsymbol{v}_2 = (1, 1, 1, 1)$,
c. $\boldsymbol{v}_3 = (-4, 6, -13, 4)$.

Question 1: Solution

View each vector as a column vector, and let

$$\boldsymbol{A} = \left(\begin{array}{ccc|c|c|c} \boldsymbol{u}_1 & \boldsymbol{u}_2 & \boldsymbol{u}_3 & \boldsymbol{v}_1 & \boldsymbol{v}_2 & \boldsymbol{v}_3 \end{array}\right) = \left(\begin{array}{ccc|c|c|c} 2 & 3 & -1 & 2 & 1 & -4 \\ 1 & -1 & 0 & 3 & 1 & 6 \\ 0 & 5 & 2 & -7 & 1 & -13 \\ 3 & 2 & 1 & 3 & 1 & 4 \end{array}\right).$$

$$\mathrm{RREF}(\boldsymbol{A}) = \left(\begin{array}{ccc|c|c|c} 1 & 0 & 0 & 2 & 0 & 3 \\ 0 & 1 & 0 & -1 & 0 & -3 \\ 0 & 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{array}\right).$$

Thus,

$$\boldsymbol{v}_1 = 2\boldsymbol{u}_1 - \boldsymbol{u}_2 - \boldsymbol{u}_3 \in \mathrm{span}\{\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3\};$$
$$\boldsymbol{v}_3 = 3\boldsymbol{u}_1 - 3\boldsymbol{u}_2 + \boldsymbol{u}_3 \in \mathrm{span}\{\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3\};$$
$$\boldsymbol{v}_2 \notin \mathrm{span}\{\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3\}.$$

Question 1: Solution

View each vector as a column vector, and let

$$\boldsymbol{A} = \left(\begin{array}{ccc|c|c|c} \boldsymbol{u}_1 & \boldsymbol{u}_2 & \boldsymbol{u}_3 & \boldsymbol{v}_1 & \boldsymbol{v}_2 & \boldsymbol{v}_3 \end{array}\right) = \left(\begin{array}{ccc|c|c|c} 2 & 3 & -1 & 2 & 1 & -4 \\ 1 & -1 & 0 & 3 & 1 & 6 \\ 0 & 5 & 2 & -7 & 1 & -13 \\ 3 & 2 & 1 & 3 & 1 & 4 \end{array}\right).$$

$$\text{RREF}(\boldsymbol{A}) = \left(\begin{array}{ccc|c|c|c} 1 & 0 & 0 & 2 & 0 & 3 \\ 0 & 1 & 0 & -1 & 0 & -3 \\ 0 & 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{array}\right).$$

Thus,

$$\boldsymbol{v}_1 = 2\boldsymbol{u}_1 - \boldsymbol{u}_2 - \boldsymbol{u}_3 \in \text{span}\{\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3\};$$
$$\boldsymbol{v}_3 = 3\boldsymbol{u}_1 - 3\boldsymbol{u}_2 + \boldsymbol{u}_3 \in \text{span}\{\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3\};$$
$$\boldsymbol{v}_2 \notin \text{span}\{\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3\}.$$

Condition for $\mathrm{span}(S) = \mathbb{R}^n$

Let $S = \{\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_k\} \subseteq \mathbb{R}^n$. View all vectors as columns vectors, and let $\boldsymbol{A} = \begin{pmatrix} \boldsymbol{u}_1 & \boldsymbol{u}_2 & \cdots & \boldsymbol{u}_k \end{pmatrix}$.
Then

$$\begin{aligned} \mathrm{span}(S) = \mathbb{R}^n &\Leftrightarrow \text{for all } v \in \mathbb{R}^n,\ \boldsymbol{A}\boldsymbol{x} = \boldsymbol{v} \text{ has a solution} \\ &\Leftrightarrow \text{for all } v \in \mathbb{R}^n,\ \boldsymbol{A}\boldsymbol{x} = \boldsymbol{v} \text{ is consistent} \\ &\Leftrightarrow \mathrm{RREF}(\boldsymbol{A}) \text{ has no zero row} \\ &\Leftrightarrow \text{all rows of } \mathrm{RREF}(\boldsymbol{A}) \text{ are nonzero} \\ &\Leftrightarrow \text{all rows of any row echelon form of } \boldsymbol{A} \text{ are nonzero.} \end{aligned}$$

Theorem

If $k < n$, then $\mathrm{span}(S) \neq \mathbb{R}^n$.

Question 2

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Determine whether each of the following sets spans $\mathbb{R}^4$:

a. $S_1 = \{(1,0,0,1),(0,1,0,0),(1,1,1,1),(1,1,1,0)\}$.

b. $S_2 = \{(1,2,1,0),(1,1,-1,0),(0,0,0,1)\}$.

c. $S_3 = \{(6,4,-2,4),(2,0,0,1),(3,2,-1,2),(5,6,-3,2),(0,4,-2,-1)\}$.

d. $S_4 = \{(1,1,0,0),(1,2,-1,1),(0,0,1,1),(2,1,2,1),(1,2,3,4)\}$.

Question 2: Solution

a. $\begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix} \xrightarrow[\text{Jordan}]{\text{Gauss-}} \boldsymbol{I}_4$. Thus S_1 spans $\mathbb{R}^4$.

b. S_2 has only 3 vectors, so S_2 does not span $\mathbb{R}^4$.

c. $\begin{pmatrix} 6 & 2 & 3 & 5 & 0 \\ 4 & 0 & 2 & 6 & 4 \\ -2 & 0 & -1 & -3 & -2 \\ 4 & 1 & 2 & 2 & -1 \end{pmatrix} \xrightarrow[\text{elimination}]{\text{Gaussian}} \begin{pmatrix} 6 & 2 & 3 & 5 & 0 \\ 0 & -\frac{4}{3} & 0 & \frac{8}{3} & 4 \\ 0 & 0 & 0 & -2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$. Thus S_3 does not span $\mathbb{R}^4$.

d. $\begin{pmatrix} 1 & 1 & 0 & 2 & 1 \\ 1 & 2 & 0 & 1 & 2 \\ 0 & -1 & 1 & 2 & 3 \\ 0 & 1 & 1 & 1 & 4 \end{pmatrix} \xrightarrow[\text{elimination}]{\text{Gaussian}} \begin{pmatrix} 1 & 1 & 0 & 2 & 1 \\ 0 & 1 & 0 & -1 & 1 \\ 0 & 0 & 1 & 1 & 4 \\ 0 & 0 & 0 & 1 & -1 \end{pmatrix}$. Thus S_4 spans $\mathbb{R}^4$.

Question 2: Solution

a. $\begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix} \xrightarrow[\text{Jordan}]{\text{Gauss-}} \boldsymbol{I}_4$. Thus S_1 spans $\mathbb{R}^4$.

b. S_2 has only 3 vectors, so S_2 does not span $\mathbb{R}^4$.

c. $\begin{pmatrix} 6 & 2 & 3 & 5 & 0 \\ 4 & 0 & 2 & 6 & 4 \\ -2 & 0 & -1 & -3 & -2 \\ 4 & 1 & 2 & 2 & -1 \end{pmatrix} \xrightarrow[\text{elimination}]{\text{Gaussian}} \begin{pmatrix} 6 & 2 & 3 & 5 & 0 \\ 0 & -\frac{4}{3} & 0 & \frac{8}{3} & 4 \\ 0 & 0 & 0 & -2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$. Thus S_3 does not span $\mathbb{R}^4$.

d. $\begin{pmatrix} 1 & 1 & 0 & 2 & 1 \\ 1 & 2 & 0 & 1 & 2 \\ 0 & -1 & 1 & 2 & 3 \\ 0 & 1 & 1 & 1 & 4 \end{pmatrix} \xrightarrow[\text{elimination}]{\text{Gaussian}} \begin{pmatrix} 1 & 1 & 0 & 2 & 1 \\ 0 & 1 & 0 & -1 & 1 \\ 0 & 0 & 1 & 1 & 4 \\ 0 & 0 & 0 & 1 & -1 \end{pmatrix}$. Thus S_4 spans $\mathbb{R}^4$.

Question 2: Solution

a. $\begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix} \xrightarrow[\text{Jordan}]{\text{Gauss-}} \boldsymbol{I}_4$. Thus S_1 spans $\mathbb{R}^4$.

b. S_2 has only 3 vectors, so S_2 does not span $\mathbb{R}^4$.

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d. $\begin{pmatrix} 1 & 1 & 0 & 2 & 1 \\ 1 & 2 & 0 & 1 & 2 \\ 0 & -1 & 1 & 2 & 3 \\ 0 & 1 & 1 & 1 & 4 \end{pmatrix} \xrightarrow[\text{elimination}]{\text{Gaussian}} \begin{pmatrix} 1 & 1 & 0 & 2 & 1 \\ 0 & 1 & 0 & -1 & 1 \\ 0 & 0 & 1 & 1 & 4 \\ 0 & 0 & 0 & 1 & -1 \end{pmatrix}$. Thus S_4 spans $\mathbb{R}^4$.

Question 2: Solution

(a) $\begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix} \xrightarrow[\text{Jordan}]{\text{Gauss-}} \boldsymbol{I}_4$. Thus S_1 spans $\mathbb{R}^4$.

(b) S_2 has only 3 vectors, so S_2 does not span $\mathbb{R}^4$.

(c) $\begin{pmatrix} 6 & 2 & 3 & 5 & 0 \\ 4 & 0 & 2 & 6 & 4 \\ -2 & 0 & -1 & -3 & -2 \\ 4 & 1 & 2 & 2 & -1 \end{pmatrix} \xrightarrow[\text{elimination}]{\text{Gaussian}} \begin{pmatrix} 6 & 2 & 3 & 5 & 0 \\ 0 & -\frac{4}{3} & 0 & \frac{8}{3} & 4 \\ 0 & 0 & 0 & -2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$. Thus S_3 does not span $\mathbb{R}^4$.

(d) $\begin{pmatrix} 1 & 1 & 0 & 2 & 1 \\ 1 & 2 & 0 & 1 & 2 \\ 0 & -1 & 1 & 2 & 3 \\ 0 & 1 & 1 & 1 & 4 \end{pmatrix} \xrightarrow[\text{elimination}]{\text{Gaussian}} \begin{pmatrix} 1 & 1 & 0 & 2 & 1 \\ 0 & 1 & 0 & -1 & 1 \\ 0 & 0 & 1 & 1 & 4 \\ 0 & 0 & 0 & 1 & -1 \end{pmatrix}$. Thus S_4 spans $\mathbb{R}^4$.

Condition for $\mathrm{span}(S) \subseteq \mathrm{span}(T)$

Let $S = \{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_k\}$ and $T = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_l\}$ be subsets of $\mathbb{R}^n$.
View all vectors as column vectors. Let $\boldsymbol{A}_S$ be the $n \times k$ matrix whose columns are vectors in S, and let $\boldsymbol{A}_T$ be the $n \times l$ matrix whose columns are vectors in T.
Then

$$\begin{aligned} \mathrm{span}(S) \subseteq \mathrm{span}(T) &\Leftrightarrow S \subseteq \mathrm{span}(T) \\ &\Leftrightarrow \text{all pivot columns of } \mathrm{REF}\left(\begin{array}{c|c} \boldsymbol{A}_T & \boldsymbol{A}_S \end{array}\right) \\ &\quad \text{are among the leftmost } l \text{ columns.} \end{aligned}$$

Question 3

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Let $\boldsymbol{u}_1 = (2, -2, 0)$, $\boldsymbol{u}_2 = (-1, 1, -1)$, $\boldsymbol{u}_3 = (0, 0, 9)$, $\boldsymbol{v}_1 = (1, -1, -5)$, $\boldsymbol{v}_2 = (0, 1, 1)$. Determine whether

a. $\text{span}\{\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3\} \subseteq \text{span}\{\boldsymbol{v}_1, \boldsymbol{v}_2\}$.

b. $\text{span}\{\boldsymbol{v}_1, \boldsymbol{v}_2\} \subseteq \text{span}\{\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3\}$.

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b. $\text{span}\{\boldsymbol{v}_1, \boldsymbol{v}_2\} \subseteq \text{span}\{\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3\}$.

Solution.

a. $\left(\begin{array}{cc|ccc} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \boldsymbol{u}_1 & \boldsymbol{u}_2 & \boldsymbol{u}_3 \end{array}\right) \xrightarrow[\text{Elimination}]{\text{Gaussian}} \left(\begin{array}{cc|ccc} 1 & 0 & 2 & -1 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 10 & -6 & 9 \end{array}\right)$. Thus $\text{span}\{\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3\} \not\subseteq \text{span}\{\boldsymbol{v}_1, \boldsymbol{v}_2\}$.

Question 3

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Solution.

a. $\left(\begin{array}{cc|ccc} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \boldsymbol{u}_1 & \boldsymbol{u}_2 & \boldsymbol{u}_3 \end{array}\right) \xrightarrow[\text{Elimination}]{\text{Gaussian}} \left(\begin{array}{cc|ccc} 1 & 0 & 2 & -1 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 10 & -6 & 9 \end{array}\right)$. Thus

$\operatorname{span}\{\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3\} \not\subseteq \operatorname{span}\{\boldsymbol{v}_1, \boldsymbol{v}_2\}$.

b. $\left(\begin{array}{ccc|cc} \boldsymbol{u}_1 & \boldsymbol{u}_2 & \boldsymbol{u}_3 & \boldsymbol{v}_1 & \boldsymbol{v}_2 \end{array}\right) \xrightarrow[\text{Elimination}]{\text{Gaussian}} \left(\begin{array}{ccc|cc} 2 & -1 & 0 & 1 & 0 \\ 0 & -1 & 9 & -5 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right)$. Thus

$\operatorname{span}\{\boldsymbol{v}_1, \boldsymbol{v}_2\} \not\subseteq \operatorname{span}\{\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3\}$.

Question 4

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Let $\boldsymbol{u}_1 = (1, 6, 4)$, $\boldsymbol{u}_2 = (2, 4, -1)$, $\boldsymbol{u}_3 = (-1, 2, 5)$, $\boldsymbol{v}_1 = (1, -2, -5)$, $\boldsymbol{v}_2 = (0, 8, 9)$. Determine whether

a. $\text{span}\{\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3\} \subseteq \text{span}\{\boldsymbol{v}_1, \boldsymbol{v}_2\}$.

b. $\text{span}\{\boldsymbol{v}_1, \boldsymbol{v}_2\} \subseteq \text{span}\{\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3\}$.

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Solution.

a. $\left(\begin{array}{cc|ccc} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \boldsymbol{u}_1 & \boldsymbol{u}_2 & \boldsymbol{u}_3 \end{array}\right) \xrightarrow[\text{Elimination}]{\text{Gaussian}} \left(\begin{array}{cc|ccc} 1 & 0 & 1 & 2 & -1 \\ 0 & 8 & 8 & 8 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$. Thus $\operatorname{span}\{\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3\} \subseteq \operatorname{span}\{\boldsymbol{v}_1, \boldsymbol{v}_2\}$.

Question 4

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Let $\boldsymbol{u}_1 = (1, 6, 4)$, $\boldsymbol{u}_2 = (2, 4, -1)$, $\boldsymbol{u}_3 = (-1, 2, 5)$, $\boldsymbol{v}_1 = (1, -2, -5)$, $\boldsymbol{v}_2 = (0, 8, 9)$. Determine whether

a. $\operatorname{span}\{\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3\} \subseteq \operatorname{span}\{\boldsymbol{v}_1, \boldsymbol{v}_2\}$.

b. $\operatorname{span}\{\boldsymbol{v}_1, \boldsymbol{v}_2\} \subseteq \operatorname{span}\{\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3\}$.

Solution.

a. $\left(\begin{array}{cc|ccc} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \boldsymbol{u}_1 & \boldsymbol{u}_2 & \boldsymbol{u}_3 \end{array}\right) \xrightarrow[\text{Elimination}]{\text{Gaussian}} \left(\begin{array}{cc|ccc} 1 & 0 & 1 & 2 & -1 \\ 0 & 8 & 8 & 8 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$. Thus $\operatorname{span}\{\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3\} \subseteq \operatorname{span}\{\boldsymbol{v}_1, \boldsymbol{v}_2\}$.

b. $\left(\begin{array}{ccc|cc} \boldsymbol{u}_1 & \boldsymbol{u}_2 & \boldsymbol{u}_3 & \boldsymbol{v}_1 & \boldsymbol{v}_2 \end{array}\right) \xrightarrow[\text{Elimination}]{\text{Gaussian}} \left(\begin{array}{ccc|cc} 1 & 2 & -1 & 1 & 0 \\ 0 & -8 & 8 & -8 & 8 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$.

Thus $\operatorname{span}\{\boldsymbol{v}_1, \boldsymbol{v}_2\} \subseteq \operatorname{span}\{\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3\}$.



Next Lesson ...

We will explore **Sections 3.3 and 3.4**.
Please watch the video clips for these sections before then.